



FRONTEND DEVELOPMENT

Complete Learning Roadmap

Learn browser fundamentals, modern UI engineering, and production-grade frontend performance.

HTML semantics

CSS layout systems

JavaScript essentials

Git workflows

DOM events

Async data fetching

01

Fundamentals

02

Core Concepts

03

Tools and
Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

Frontend



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

HTML semantics • CSS layout systems • JavaScript essentials • Git workflows

02

Core Concepts

DOM events • Async data fetching • Accessibility basics • Responsive design

03

Tools and Frameworks

TypeScript • React

04

Real Projects

Design system page • Dashboard with API data • Auth-protected app • Testing workflow

05

Advanced Topics

Performance budgets • Web Vitals optimization • Architecture scaling • Observability for UX

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- HTML semantics
- CSS layout systems
- JavaScript essentials
- Git workflows

Tools & Frameworks

- HTML semantics
- CSS layout systems
- JavaScript essentials
- Git workflows

Recommended Projects

- Frontend Development workshop using HTML semantics
- Frontend Development portfolio project with CSS layout systems

Stage 1 Milestone Complete Frontend Development workshop using HTML semantics and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- DOM events
- Async data fetching
- Accessibility basics
- Responsive design

Tools & Frameworks

- DOM events
- Async data fetching
- Accessibility basics
- Responsive design

Recommended Projects

- Frontend Development workshop using DOM events
- Frontend Development portfolio project with Async data fetching

Stage 2 Milestone Complete Frontend Development workshop using DOM events and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- TypeScript
- React
- Routing patterns
- State management

Tools & Frameworks

- TypeScript
- React

Recommended Projects

- Frontend Development workshop using TypeScript
- Frontend Development portfolio project with React

Stage 3 Milestone Complete Frontend Development workshop using TypeScript and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Design system page
- Dashboard with API data
- Auth-protected app
- Testing workflow

Tools & Frameworks

- Design system page
- Dashboard with API data
- Auth-protected app
- Testing workflow

Recommended Projects

- Frontend Development workshop using Design system page
- Frontend Development portfolio project with Dashboard with API data

Stage 4 Milestone Complete Frontend Development workshop using Design system page and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

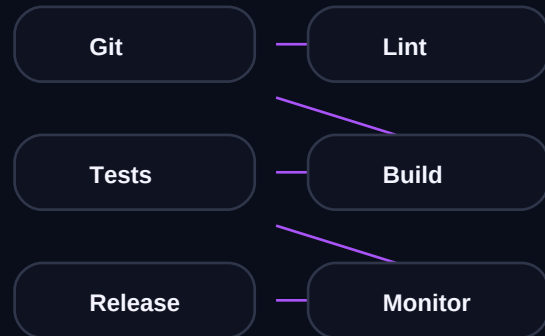
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Performance budgets
- Web Vitals optimization
- Architecture scaling
- Observability for UX

CI/CD Pipeline



Stage 5 Milestone

A production-grade frontend development workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Frontend Development docs and community guides

Link: <https://developer.mozilla.org/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/frontend>

PAID

Structured specialization track focused on Frontend Development

Link: <https://roadmap.sh/frontend>

FREE

Frontend Development roadmap reference

Link: <https://roadmap.sh/frontend>

FREE

Official documentation

Link: <https://developer.mozilla.org/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Frontend Development?
- Which project best proves Stage 4 competency in Frontend Development?
- What trade-offs appear in Stage 5 decisions for Frontend Development?